

# **Business Model Report on ASML Holding N.V.**

Savith Rallapalli  
B.Tech- Industrial and Systems Engineering  
Indian Institute of Technology Kharagpur

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## **Business Model Summary**

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### **Introduction**

ASML's business model is based upon selling advanced lithography systems to semiconductor fabs. Lithography systems (such as EUV, and DUV) are a vital component of the semiconductor manufacturing process. ASML's main customers include NVIDIA, Apple, TSMC, Samsung, and Intel, all of which are major players in the semiconductor industry.

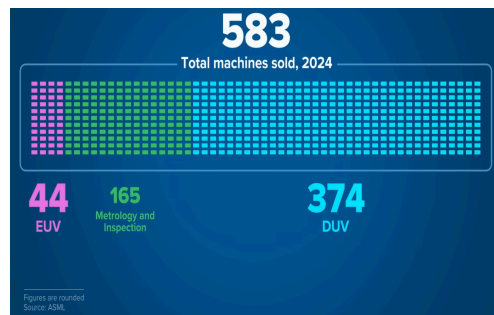
Lithography turns out to be one of the major bottlenecks in the advanced semiconductor value chain, as this process is a crucial step in the production process. Issues in this part of the value chain would lead to years of delays in semiconductor advancements and A.I. hardware scaling.

The objective of this report is to explain ASML's business strategy, where they stand in this market, and what key risks and opportunities are present.

### **Business Model Overview**

ASML produces extreme ultraviolet (EUV) lithography machines and deep ultraviolet (DUV) lithography machines, which enables leading edge node manufacturing. These lithography systems enable the manufacturing of smaller, more advanced nanochips. ASML's machines allow its customers to stay at the forefront of their market.

ASML generates revenue through its sale of these lithography systems to its customers and also through additional services, like the maintenance of systems, system upgrades, support contracts, &c.



Specific to ASML's EUV systems, their next generation technology of High NA (Numerical Aperture) is actively in production and has recently begun delivering these systems to customers. ASML still sells its DUV systems, and they are a majority of its machine sales. ASML also sells some complementary metrology and other manufacturing process equipment systems, though contributing very minimally to sales.

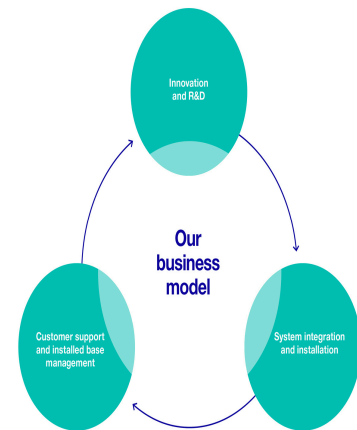
Supplier relations being a priority to them, ASML maintains a very tight knit connection with its suppliers. ASML brings on suppliers very early on during product development cycles. A key supplier for the ultra-flat mirrors necessary in these lithography systems is Carl Zeiss SMT, which ASML collaborates closely with to co-develop technology.

The business model is distantly related to the razor blade model, in which it sells the lithography systems for hundreds of millions of euros, and continuously gets further revenue from upgradation and maintenance checks done on these lithography systems, accounting for a stable 25-30% of their revenue. But it's also distinct from the razor blade model in the sense that the initial purchase by customers here also accounts for a bulk of the sales. The customers of ASML are locked in with them due to the engineering complexity, patented technology and large capital requirements that prevent any other company from producing similar systems in a strategic time frame.

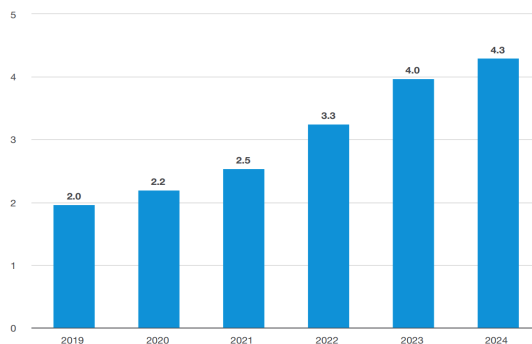
## Value Chain & Operations

The three key things that stand out in ASML's value chain are:

Firstly, a specialized supplier ecosystem is a key part of ASML's value chain. ASML sources its components for its lithography systems from thousands of suppliers, mainly owing to the fact that the parts required in these lithography systems are highly complex and more often than not only have one key supplier globally. A few key examples of this are, Carl Zeiss SMT for the ultra-flat optical mirrors as stated before and Cymer being the supplier of the plasma light technology for the lithography systems (which has now been acquired by ASML). The operational stability of this megacorporation lies in its ultra-specialization of partners in its supply chain.



R&D investments (costs) in € billion



Another important part of ASML's value chain is its R&D presence as a part of its operations. As shown to the left, ASML is spending heavily and increasing this investment year by year in the research of the complex physics and engineering behind their \$400 million dollar systems. R&D is a core operation for ASML, and accounts for roughly 15% of its revenue, rather than being a side support operation.

Lastly, a strong customer relationship is vital to ASML's operational philosophy. ASML is heavily involved with their customers throughout the process of procurement and installation of the lithography systems, catering to the needs of the semiconductor manufacturer personally. ASML boasts about their customer service availability globally, and how they have a long legacy of trust with their customer base.

## Market Positioning & Competitive Landscape

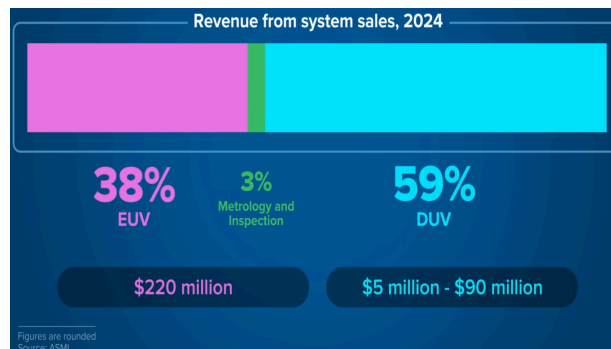
ASML stands at a very unique position in its market and amongst its competitors. It holds a unique position by completely dominating its market and leaves a minimal addressable market for its competitors. ASML is the sole supplier of EUV lithography systems globally, and is the major player in the DUV lithography segment. What ASML has is essentially a monopoly in the EUV segment, and a near monopoly in the DUV segment.

In the DUV lithography market, ASML's primary competitors are Canon and Nikon, which remain active in the legacy and mid-range DUV market, whereas ASML dominates in the sub 45nm segment. Canon and Nikon have tried to develop competing technologies especially for EUV, but due to prohibitive R&D CAPEX requirements they have failed.

ASML also sells tools for metrology and other processes, which means it's competing with other semiconductor equipment companies like Applied Materials and Lam Research for market share in that segment. The competitive landscape looks very promising for ASML. After decades of heavy investing back into their R&D, they have fortified their dominant position in this market and made sure that no one else can beat them without heavy and very restrictive CAPEX into the research and development of these complex systems. Few challengers exist right now globally, besides China, which is attempting to make its own domestically manufactured lithography systems.

## Finance Model & Performance

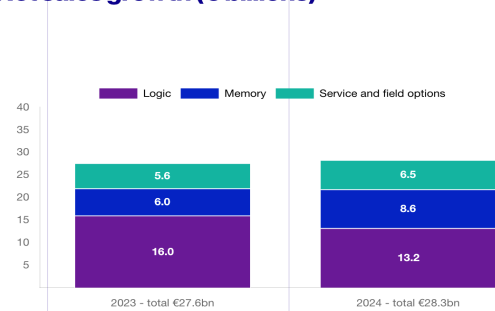
ASML's business model translates into a revenue mix of EUV/DUV sales and system installation & service sales.



ASML has 3 main revenue streams aligning with its 3 main components of its value chain. From system sales specifically, EUV contributes majorly even though it is a minority in terms of sales volume. Metrology and inspection tools account for only 3% of sales and another source of revenue is the installed base management (recurring services, upgrades, &c.).

Total net sales have grown from 27.6 billion euros in 2023 to 28.3 billion euros in 2024. Though this was seen with a decrease in net system sales from 22 billion euros in 2023 to 21.8 billion euros in 2024. Net service and field options sales increased from 2023 from 5.6 billion euros to 6.5 billion euros. In terms of profitability gross profit hovered at around the same figure of 51% percent compared to last year.

### Net sales growth (€ billions)



In terms of liquidity, ASML seems to be pretty healthy, with around 9 billion euros of free cash flow, a large increase from 3 billion euros last year.

## Strategic Risk & Opportunities

ASML's success is highly dependent on the protection of their intellectual property ecosystem. ASML and their partners (customers) have reported cyber- and other security threats. This poses a high risk to their operations, and their stability as the dominant lithography system provider.

Geopolitical tensions and the global strive for technological sovereignty are rising. This may disrupt ASML's operations, as there is a risk in future trade restrictions owing to these tensions. These restrictions will disrupt ASML in sourcing their parts, selling their systems, and servicing them to certain customers. This risk factor is increasing as the race for AI computational supremacy is adding to the already existing geopolitical tensions.

ASML may also face scaling issues due to the exponential rise in demand for advanced hardware. ASML states they may face issues in increasing production capacity throughout their supply chain due to rising demand, and onboarding more and more well trained engineers who can deal with the extremely complex physics behind these systems is a challenge. Climate change and resource scarcity also pose a threat to ASML in sourcing their materials for the production of their systems.

Due to AI scaling ASML has the possibility of profiting greatly off the exponential demand for advanced nodes. Also if high-NA EUV adoption increases, ASML could increase their net income largely due to the increased price range of the technologies.

## **Conclusion**

ASML operates one of the most technologically complex and important business models in the semiconductor ecosystem. Its vertically integrated R&D, strong supplier relations, state of the art technology, and monopoly in EUV lithography, create a nearly indestructible value chain. While its supply chain dependence, geopolitical tensions, IP protection dependence, and possible scaling issues pose real risks for the company, ASML's long term expansion remains strong, driven by AI scaling, migration towards advanced node technologies, and global semiconductor capacity expansion.

## **References**

*Business Model Report on ASML Holding N.V.*

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